

Team Members:

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Introduction:

This project aims to develop an explainable hybrid recommender system to assist researchers in discovering relevant academic papers, emphasizing transparency and interpretability over black-box models.

Dataset:

ArXiv Metadata Dataset (from Kaggle): includes titles, abstracts, authors, categories, and publication history. A representative subset will be sampled using streaming techniques to ensure efficient processing and diversity across domains.

Ratings:

The system will predict implicit feedback (e.g., reading, saving, citing) by generating a relevance score (0–1) based on content similarity, user interactions, and popularity signals.

Techniques

- Content-Based Filtering: TF-IDF vectorization of titles and abstracts, cosine similarity for semantic relatedness
- User Profiling: aggregation of user interaction history to infer research interests
- Popularity-Based Re-ranking: incorporating interaction frequency to balance relevance and popularity
- Hybrid Model: weighted score combining content similarity and popularity
- Explainability Layer: providing overlapping keywords, contribution analysis, and simple natural language explanations for each recommendation

Undertaking:

We pledge to conduct this project with full academic integrity, avoiding any dishonest means throughout development, experimentation, and reporting.